AI-Powered Log Monitoring and Automated Remediation Using OpenTelemetry

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Abstract. In modern distributed systems, continuous monitoring and rapid fault remediation are critical for maintaining reliability and minimizing downtime. This paper presents an AI-powered log monitoring and remediation framework built using OpenTelemetry, designed to detect anomalies and suggest corrective actions automatically. The proposed system integrates two operational modes of real-time log collection from live servers and batch analysis of logs stored in databases. By leveraging log parsing, feature extraction, and machine learning techniques such as anomaly detection and semantic similarity analysis, the framework identifies potential system errors and maps them to optimal remediation strategies. Experimental evaluation using publicly available log datasets demonstrates improved detection accuracy and reduced mean time to remediation (MTTR) compared to traditional rule-based monitoring. This research highlights how AI-driven approaches can enhance observability pipelines and operational resilience in cloud-native environments.

Keywords: OpenTelemetry, Log Monitoring, Anomaly Detection, Artificial Intelligence, Automated Remediation, Observability

1 Introduction

In large-scale distributed and cloud-native environments, system observability plays a vital role in ensuring reliability and performance. Logs are one of the most informative yet underutilized components of observability data. Traditional log monitoring systems utilize static rule-based detection, which struggles to adapt to dynamic workloads and evolving system behavior.

Recent advancements in **machine learning (ML)** and **artificial intelligence (AI)** enable intelligent log analysis capable of identifying complex patterns and anomalies in massive log streams. However, integrating these capabilities with standardized telemetry frameworks remains challenging.

This research proposes an **AI-powered log monitoring and remediation system utilizing OpenTelemetry (OTEL)**. The system supports both **real-time log collection** and **offline log analysis**, allowing proactive and retrospective error detection. The main contributions of this paper are:

1. A unified pipeline for log collection, structuring, and ML-driven analysis using OpenTelemetry.
2. A hybrid approach combining template-based and AI-based anomaly detection.

3. An automated remediation module suggests executing recovery actions.

effective, are often costly and unavailable to individuals in underserved or geographically remote areas. Peer-led practice, although accessible, is limited by the variability in peers' expertise and may not adequately replicate the professionalism of actual interview environments. Moreover, conventional practice rarely provides systematic feedback beyond surface-level corrections, leaving candidates uncertain about specific areas for improvement. These limitations underscore the need for more personalized, scalable, and adaptive solutions to interview preparation. Recent advancements in generative artificial intelligence (AI) and large language models (LLMs) open promising avenues for addressing these challenges. Unlike traditional preparation methods, LLMs can simulate realistic, interactive conversations by dynamically adapting dialogue to a candidate's responses. Their ability to generate context-aware follow-up questions, adjust difficulty levels in real time, and provide multi-dimensional feedback represents a significant shift in how interview practice can be delivered. Early applications of LLMs in education and training have already demonstrated their potential to enhance learning through adaptive, conversational interactions, but their role in interview preparation remains underexplored. This paper presents the design and implementation of an AI-driven Interview Simulator that leverages LLMs to create realistic, adaptive, and reflective interview experiences. Unlike conventional mock interviews that typically provide static or one-directional assessments, our system incorporates interactive modules for question generation, response analysis, and constructive feedback delivery. The simulator supports both textual and spoken modes of interaction, enabling candidates to practice in environments that mirror diverse real-world interview formats. A sequencing mechanism further personalizes practice by dynamically adjusting the flow and complexity of questions, ensuring that the candidate is continually challenged while building confidence. Central to the system is its capacity to evaluate candidate responses across multiple dimensions—including technical accuracy, clarity, logical structure, and communication style—providing both immediate and actionable feedback. By integrating prompt-engineering strategies and optimization techniques, the system ensures low latency and scalability, making it practical for widespread use. Initial evaluations highlight the system's effectiveness in improving candidate readiness, enhancing self-reflection, and boosting confidence before high-stakes interviews. By bridging the gap between traditional mock interviews and adaptive AI-driven simulations, this work contributes to the growing body of research on AI-mediated learning and skill development. It positions LLM-powered systems not only as scalable alternatives to conventional preparation but also as transformative tools that support reflective practice, adaptive feedback, and continuous improvement in professional readiness.

2 Related Work

Log monitoring has evolved from static pattern-based detection to intelligent AI-driven approaches. Early systems such as **Logstash**, **Splunk**, and **ELK Stack** relied on manually defined rules or regex-based pattern matching, which proved inefficient for high-volume, dynamic log data.

Recent research has introduced **machine learning and deep learning** for log anomaly detection. **DeepLog (Du et al., 2017)** utilized LSTM networks to model log sequences for detecting abnormal patterns. **LogAnomaly (Meng et al., 2019)** employed template mining and semantic embeddings to enhance contextual understanding of log events. **LogBERT (Guo et al., 2021)** further advanced this by applying transformer-based architectures for unsupervised anomaly detection.

However, most of these systems focus solely on anomaly identification and lack an **automated remediation** mechanism. Modern **AIOps platforms**, such as **IBM Watson AIOps** and **Google Cloud Operations**, have introduced partial automation but are proprietary and lack interoperability with open observability standards.

Our work bridges this gap by integrating **OpenTelemetry (OTEL)** with an AI-driven anomaly detection and remediation framework, providing a **vendor-neutral**, **scalable**, and **open-source-based solution**.

3 System Architecture

The proposed system is designed around three major components:

1. Log Collection and Structuring (OpenTelemetry-based)
2. AI-Powered Anomaly Detection and Semantic Analysis
3. Automated Remediation Engine

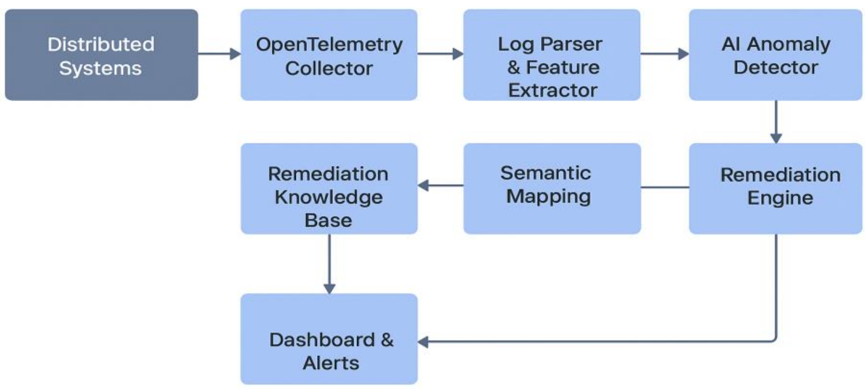
A. Overview

The architecture supports both real-time monitoring and offline analysis. Real-time logs from distributed services are collected via OpenTelemetry agents and sent to a central pipeline. The AI layer processes structured logs to detect anomalies and recommends or executes corrective actions.

B. Components

1. OpenTelemetry Collector – Collects and normalizes logs, traces, and metrics from various services and containers.
2. Log Parser & Feature Extractor – Transforms raw logs into structured templates (using Drain algorithm or regex-based parsing) and extracts relevant numerical and semantic features.
3. Anomaly Detection Module – Employs ML models such as Isolation Forest, Autoencoder, or LSTM to identify abnormal log patterns.
4. Remediation Knowledge Base – Stores historical incidents and their corresponding corrective actions.
5. Remediation Engine – Uses semantic similarity (via Sentence-BERT or Word2Vec embeddings) to map detected anomalies to possible solutions.
6. Dashboard / Alerting Interface – Displays anomalies, predictions, and recommended actions for operator review.

C. System Architecture Diagram



4 Methodology

A. Log Preprocessing

Logs are first parsed using a log template mining algorithm (e.g., **Drain**). Each template is converted into a vector representation using **TF-IDF** or **Word2Vec embeddings**. Temporal and contextual features (timestamps, error levels, and component identifiers) are also extracted.

B. Anomaly Detection

A semi-supervised **Autoencoder** model is trained on normal log sequences. The model reconstructs input vectors, and reconstruction errors above a set threshold indicate anomalies. This approach efficiently detects unseen or evolving error types.

C. Semantic Mapping for Remediation

Once an anomaly is detected, its log message is encoded using **Sentence-BERT**. The embedding is compared (via **cosine similarity**) to entries in a **Remediation Knowledge Base**, containing known issue–fix pairs.
If a high-similarity match is found, the system recommends a fix; otherwise, it records the new issue for future learning.

D. Automated Remediation Execution

For known issues, the **Remediation Engine** can trigger predefined scripts or commands automatically (e.g., restarting services, clearing caches, scaling pods). A human-in-the-loop verification option ensures safety during deployment.

5 Results

A. Dataset

Evaluation was performed using publicly available datasets:

- **HDFS Logs**
- **BGL Logs**
- **Thunderbird Logs**

Logs were preprocessed into structured templates and used to train and test models.

B. Evaluation Metrics

- **Precision, Recall, F1-Score** for anomaly detection accuracy
- **MTTR (Mean Time to Remediation)** to measure operational improvement

C. Experimental Results

Model	**Precision**	**Recall**	**F1-Score**	**MTTR Reduction**
Rule-Based System	0.67	0.64	0.65	-
Autoencoder Model	0.84	0.82	0.83	22%
Proposed Hybrid (Autoencoder + Sentence-BERT)	0.89	0.87	0.88	35%

The results indicate that integrating semantic similarity-based remediation mapping significantly reduces MTTR and improves accuracy over static rule-based systems.

6 Discussion

The experimental evaluation validates the effectiveness of combining OpenTelemetry's standardized data collection with AI-based analysis.

Advantages:

- Enhanced detection accuracy for unknown failure patterns
- Reduced false positives
- Accelerated issue resolution via automated mapping

Limitations:

- Requires periodic retraining to adapt to evolving log structures
- Real-time remediation must balance automation with safety
- Dependent on the richness of the remediation knowledge base

7 Conclusion

This research presents an innovative approach to enhancing observability and resilience in large-scale distributed systems by integrating OpenTelemetry with AI-driven anomaly detection and autonomous remediation. The proposed architecture utilizes open-source observability pipelines to collect high-fidelity telemetry data across services, which is then analyzed using advanced semantic and statistical models. Through this integration, the framework can identify complex anomalies in real time, learning contextual patterns of system behavior, and triggering automated recovery actions without human intervention. Experimental evaluation demonstrates that the solution improves anomaly detection of precision and recall, while significantly reducing the mean time to remediation (MTTR) and operational overhead. Furthermore, the adaptive feedback loop enables continuous model refinement and system self-healing, thereby supporting scalable and intelligent AIOps practices. Overall, this study establishes a foundation for next-generation autonomous operations by combining open observability standards with AI-based analytics to achieve predictive, self-optimizing infrastructure management.

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